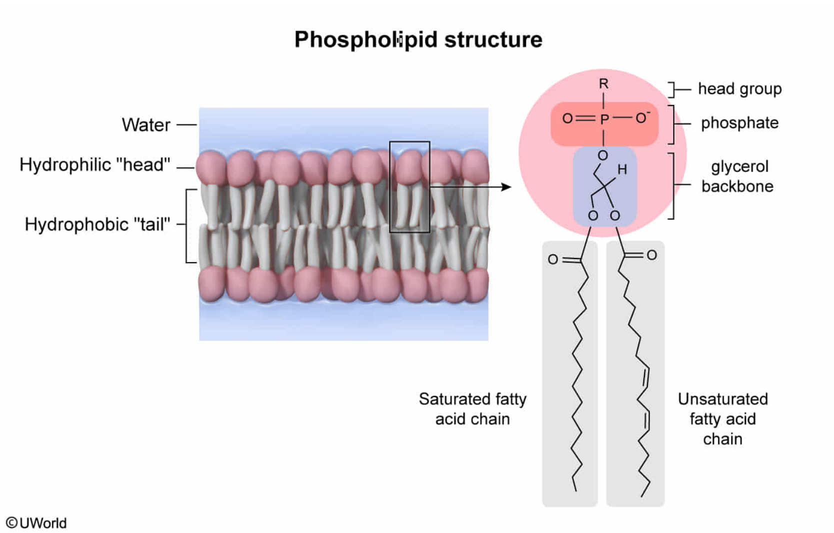
Phospholipids commonly contain all of the following structural features EXCEPT:

- A. a phosphate.
- B. a phenol. [94%]
- C. a glycerol backbone. [4%]
- D. fatty acid chains.

Correct
95% answered correctly

Explanation:



Phospholipids are a main component of the cell membrane and are responsible for the cell membrane's structure. They are amphiphilic molecules made of a hydrophobic (nonpolar) tail and a hydrophilic (polar) head group. This characteristic causes the phospholipid molecules to arrange into a bilayer in aqueous environments. The hydrophobic tails form the inner portion of the bilayer to keep away from water, and the hydrophilic heads encompass the outer portion of the bilayer toward water because the polar head group molecules can form favorable interactions with water, such as hydrogen bonds.

There are two major classes of phospholipids: glycerophospholipids and sphingophospholipids. The general structure of a phospholipid includes a polar head group and nonpolar tail joined together by a molecule referred to as the backbone. The hydrophilic head contains a substituted **phosphate (Choice A)**, and the hydrophobic tail is made of two **fatty acid chains (Choice D)**. The backbone molecule is either a **glycerol** molecule in glycerophospholipids **(Choice C)** or a sphingosine molecule for sphingophospholipids. Fatty acid chains connect to the backbone through either an ester bond (glycerophospholipids) or an amide bond (sphingophospholipids). In contrast, the phosphate group is connected to the backbone through a phosphodiester bond.

A variety of phospholipids can be made with different head groups on phosphate, a different backbone, and different fatty acids (saturated or unsaturated). The fatty acid double bond configuration causes the hydrophobic tail to take on different shapes and affects the properties of the phospholipid, such as membrane fluidity.

Like fatty acids, glycerol, and phosphates, phenols are present in biological molecules, such as the amino acid tyrosine; however, **phenols are *not* a common component** of phospholipid structure.

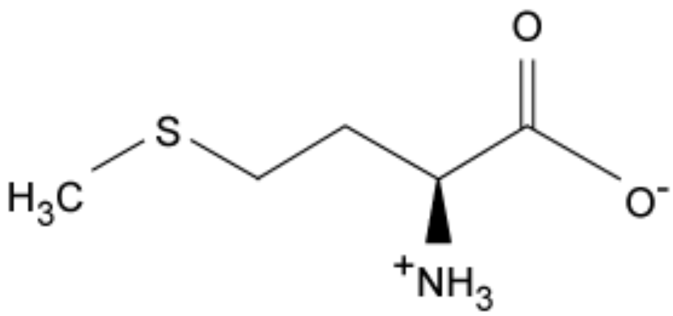
Educational objective:
Phospholipids are the main component in the cell membrane and provide structure to the cell membrane. These molecules

are made of a hydrophilic head containing a substituted phosphate group and a fatty acid hydrophobic tail. The head and tail are commonly linked together by a glycerol backbone.

Time spent:0 sec	**QID**:401768
Subject:	**System:**
Organic Chemistry	Functional Groups and Their Reactions

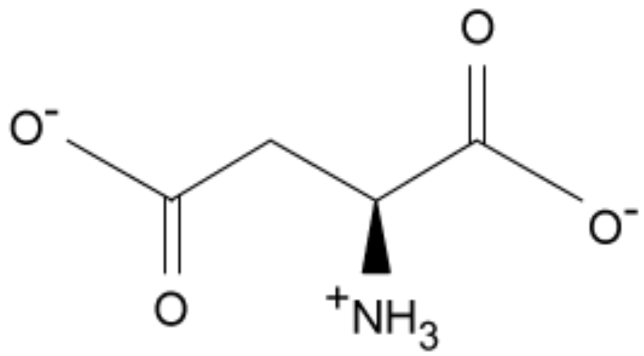
If *Staphylococcus aureus* V8 protease is specific for acidic amino acids, which of the following residues would be cleaved by this protease?

A.



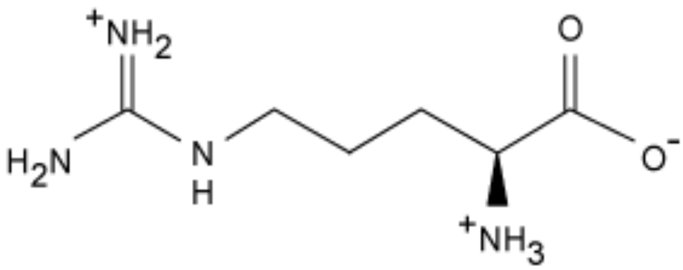
[3%]

B.



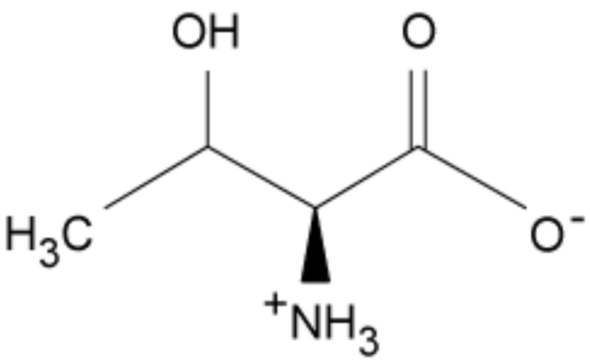
[81%]

C.



[9%]

D.

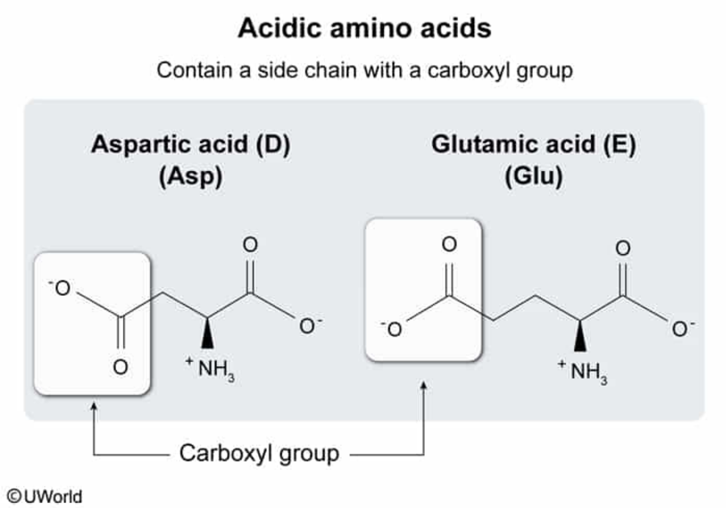


[5%]

Correct
81% answered correctly

Explanation:

微信Mai_ange



Amino acids are the fundamental components of proteins and contain three main constituents: An amino group, a carboxyl group, and a side chain, which are all bonded to one carbon atom (the α-carbon). Amino acids differ only in the structure of the **side chain**, which contains different **functional groups** with **varying characteristics**. Therefore, amino acids can be grouped together and classified based on their side chain functional groups.

Four common amino acid classifications include hydrophilic, hydrophobic, **acidic**, and basic. Acidic amino acids have a side chain with a proton-donating (acidic) functional group that is negatively charged at physiological pH. The amino group is protonated (positively charged) and the carboxyl group is deprotonated (negatively charged) at physiological pH, giving amino acids with uncharged side chains a zero net charge. Therefore, acidic amino acids have a −1 net charge at physiological pH because they have charged side chains.

Staphylococcus aureus V8 protease specifically cleaves peptide bonds at the carboxyl terminal of acidic amino acids. The distinguishing feature of acidic amino acids is the carboxyl group at the end of the side chain. The only answer choice with a carboxyl group as part of the side chain is aspartic acid.

(Choices A, C, and D) These amino acids are methionine, arginine, and threonine, which can be classified as hydrophobic, basic, and hydrophilic, respectively.

Educational objective:

Amino acids are the fundamental components of proteins and are made of an amino group, a carboxyl group, and a side chain. The structure of amino acids differs only in the side chains, which are made of different functional groups with varying characteristics. Amino acids can therefore be grouped based on their side-chain classification, including hydrophilic, hydrophobic, acidic, or basic.

Time spent: 0 sec	**QID**:401769
Subject:	**System:**
Organic Chemistry	Functional Groups and Their Reactions